Econometrics Assignment 4

Causal Inference

Assignment Overview

This assignment introduces modern causal inference methods used in economics, public policy, labor economics, development economics, and applied machine learning.

The assignment focuses on distinguishing:

- correlation
- prediction
- causal effects

You will work with observational data, natural experiments, and quasi-experimental methods.

The assignment emphasizes:

- identification
- counterfactual reasoning
- selection bias
- treatment effects
- assumptions behind causal claims

Topics to Learn Before Attempting This Assignment

A. Potential Outcomes Framework

- treatment and control
- counterfactuals
- observed vs unobserved outcomes
- causal effects
- Rubin causal model
- individual treatment effects
- average treatment effects

B. Treatment Effects

- ATE
- ATT
- ATC
- heterogeneous treatment effects
- policy relevance of ATT

C. Selection Bias and Confounding

- omitted variable bias
- selection on observables
- selection on unobservables
- confounding variables
- survivorship bias
- attrition

D. Randomized Controlled Trials

- random assignment
- internal validity
- balance
- compliance
- attrition
- treatment contamination

E. Regression Adjustment and Matching

- controlled OLS
- conditional independence
- propensity scores
- nearest-neighbor matching
- common support
- balance checking

F. Difference-in-Differences

- natural experiments
- treatment timing
- parallel trends
- two-way fixed effects
- staggered adoption intuition
- event-study intuition

G. Instrumental Variables

- endogeneity
- simultaneity
- omitted variables
- relevance
- exclusion restriction
- local average treatment effect
- weak instruments

H. Regression Discontinuity Design

- running variables
- treatment cutoffs
- local treatment effects
- continuity assumptions
- manipulation near cutoff

I. Causal Interpretation

- identification assumptions
- external validity
- robustness
- sensitivity analysis
- threats to causal inference

Required Submission

Submit:

1. One Jupyter notebook
2. One short report

For each problem, include:

- dataset description
- treatment and control definitions
- potential outcomes setup
- identification assumptions
- regression or matching model
- estimates and interpretation
- diagnostic checks
- final causal interpretation

Use:

- `statsmodels`
- `linearmodels`
- `sklearn`
- `pandas`
- `numpy`

Problem 1 — Job Training and Earnings

Dataset

Use the **LaLonde Job Training Dataset**.

This is a classic causal inference dataset analyzing whether job training programs increased later earnings.

Research Question

Did job training increase participants' earnings?

Variables

Outcome Variable

- re78 = earnings in 1978

Treatment Variable

- treat = 1 if person received training

Controls

- age
- education
- black
- hispanic
- married
- nodegree
- re74
- re75

Part A — Potential Outcomes Framework

Define:

$$Y_i(1) = \text{earnings if person } i \text{ receives training}$$

$$Y_i(0) = \text{earnings if person } i \text{ does not receive training}$$

Questions

1. What is the treatment?
2. What is the control condition?
3. What is the observed outcome?
4. What is the missing counterfactual?
5. Define ATE.
6. Define ATT.
7. Which is more relevant here: ATE or ATT? Why?

Expected intuition: policymakers usually care whether the program helped people who actually received treatment.

Part B — Naive Comparison

Estimate:

$$re78_i = \alpha + \beta treat_i + u_i$$

Tasks

1. Estimate the naive OLS regression.
2. Compare pre-treatment earnings across groups.
3. Create a balance table.

Balance Table

Variable	Treated Mean	Control Mean	Difference
age			
education			
re74			
re75			

Questions

1. Interpret β.
2. Is this estimate causal?
3. Why might treated and untreated individuals differ before training?

Part C — Regression Adjustment

Estimate:

$$
\begin{aligned}
re78_i = \alpha + \beta treat_i \\
&+ \gamma_1 age_i \\
&+ \gamma_2 education_i \\
&+ \gamma_3 black_i \\
&+ \gamma_4 hispanic_i \\
&+ \gamma_5 married_i \\
&+ \gamma_6 nodegree_i \\
&+ \gamma_7 re74_i \\
&+ \gamma_8 re75_i \\
&+ u_i
\end{aligned}
$$

Questions

1. How does β change after adding controls?
2. Which controls matter most?
3. What selection bias does regression adjustment attempt to address?
4. What selection bias may still remain?

Important: regression adjustment only controls for observed variables.

Part D — Propensity Score Matching

Estimate propensity scores:

$$P(treat = 1|X)$$

where X includes observed covariates.

Use logistic regression.

Then:

- match treated units to similar control units
- estimate ATT
- check covariate balance

Questions

1. Did matching improve balance?
2. Is common support satisfied?
3. What observations get dropped?
4. Is the ATT estimate credible?
5. How does matching compare with naive OLS?

Part E — RCT Interpretation

If using the experimental NSW sample, discuss:

- random assignment
- balance in expectation
- small-sample imbalance
- attrition
- noncompliance

Questions

1. Why does randomization help causal inference?
2. What does randomization balance in expectation?
3. Why can small samples still be imbalanced?
4. How can attrition affect validity?

Part F — Final Causal Conclusion

Write a 600–800 word report answering:

> Did job training cause higher earnings?

Your discussion must distinguish:

- naive association
- adjusted association
- matched estimate
- causal interpretation
- remaining threats

Problem 2 — Minimum Wage and Employment

Dataset

Use a Difference-in-Differences dataset based on the classic Card and Krueger fast-food minimum wage study.

Research Question

Did the minimum wage increase reduce employment in fast-food restaurants?

Concepts Covered

- natural experiments
- treatment and control groups
- Difference-in-Differences
- fixed effects
- parallel trends
- event-study intuition

Setup

Treatment Group

- New Jersey restaurants

Control Group

- Pennsylvania restaurants

Outcome Variable

- employment

Treatment Indicator

- $treated_i = 1$ if restaurant is in New Jersey

Post Indicator

- $post_t = 1$ after policy change

Part A — Potential Outcomes for DiD

Define:

$$Y_{it}(1) = \text{employment if exposed to higher minimum wage}$$

$$Y_{it}(0) = \text{employment without policy exposure}$$

Questions

1. What is the treatment?
2. What is the control group?
3. What is the missing counterfactual?
4. Why can we not simply compare New Jersey before vs after?
5. Why can we not simply compare New Jersey vs Pennsylvania after?

Part B — Manual Difference-in-Differences

Compute:

$$DiD = (NJ_{after} - NJ_{before}) - (PA_{after} - PA_{before})$$

Required Table

Group	Before Mean	After Mean	Change
New Jersey			
Pennsylvania			
Difference-in-Differences			

Questions

1. Interpret the DiD estimate.
2. Is the estimate positive or negative?
3. What would a negative estimate imply?
4. What would a positive estimate imply?

Part C — DiD Regression

Estimate:

$$employment_{it} = \alpha + \beta_1 treated_i + \beta_2 post_t + \beta_3 (treated_i \times post_t) + u_{it}$$

Interpret:

$$\beta_3 = \text{Difference-in-Differences estimate}$$

Questions

1. Does β_3 match the manual DiD estimate?
2. Is β_3 statistically significant?
3. What is the ATT interpretation?
4. Why is this not ordinary correlation?

Part D — Fixed Effects DiD

Estimate:

$$employment_{it} = \beta treated\_post_{it} + \alpha_i + \lambda_t + u_{it}$$

where:

- α_i = restaurant fixed effects
- λ_t = time fixed effects

Questions

1. What do restaurant fixed effects control for?
2. What do time fixed effects control for?
3. Why do treated and post disappear with fixed effects?
4. Does β change relative to simple DiD?

Part E — Clustered Standard Errors

Cluster standard errors at the restaurant level.

Questions

1. Why should standard errors be clustered?
2. Does statistical significance change?
3. What happens if clustering is ignored?
4. Why are repeated observations from the same restaurant not independent?

Part F — Parallel Trends Assumption

Questions

1. What is the parallel trends assumption?
2. Can it be fully tested with only one pre-period?
3. What evidence would make the assumption more plausible?
4. What would violate parallel trends?

Possible threats:

- local demand shocks
- state-specific economic conditions
- border spillovers
- anticipation effects

Part G — RDD and IV Extensions

RDD Extension

Suppose treatment applies only above or below a threshold.

Questions

1. What is the running variable?
2. What is the cutoff?
3. What is the treatment?
4. What assumption is required near the cutoff?

Core intuition: units near the cutoff should be comparable except for treatment exposure.

IV Extension

Suppose actual wage changes are endogenous.

Define:

$$Z = \text{being located in New Jersey after policy}$$

$$D = \text{actual wage increase}$$

$$Y = \text{employment}$$

Questions

1. Does Z affect wages? (relevance)
2. Does Z affect employment only through wages? (exclusion)
3. Why might exclusion fail?
4. What would IV estimate?

Part H — Final Causal Conclusion

Write a 700–900 word report answering:

> Did the minimum wage increase reduce employment?

Your answer must discuss:

- Difference-in-Differences estimate
- fixed effects estimate
- clustered standard errors
- parallel trends
- natural experiment logic
- remaining threats

Required Comparison Table

Method	Problem Used In	What It Estimates	Key Assumption	Main Weakness
Naive OLS	Job training	Association	No omitted variables	Selection bias
Controlled OLS	Job training	Conditional association	Selection on observables	Unobserved confounding
RCT	Job training	ATE / ATT	Random assignment	Attrition / non-compliance
PSM	Job training	ATT	Conditional independence	Only observed controls
DiD	Minimum wage	ATT	Parallel trends	Bad control group
Fixed Effects	Minimum wage	Within-unit effect	Time-invariant confounding only	Time-varying confounding
IV	Extension	LATE	Relevance + exclusion	Hard to justify exclusion
RDD	Extension	Local treatment effect	No manipulation at cutoff	Local estimate only

Minimum Deliverables

For each problem, submit:

1. Dataset description
2. Treatment and control definitions
3. Potential outcomes setup
4. Identification assumption
5. Main regression or model
6. Estimate table
7. Diagnostic, balance, or validity check
8. Final causal interpretation

What You Should Learn

By the end of this assignment, you should understand:

- why causal inference is difficult
- why observational correlations can be misleading
- how counterfactual reasoning works
- how treatment effects are defined
- how matching attempts to reduce selection bias
- why randomization is powerful
- how Difference-in-Differences identifies causal effects
- why fixed effects help with unobserved heterogeneity
- why IV assumptions are difficult
- why RDD identifies local causal effects

Final Reminder

Every causal estimate depends on assumptions.

You must clearly distinguish:

- association
- prediction
- causal identification

A statistically significant estimate is not automatically a credible causal effect.